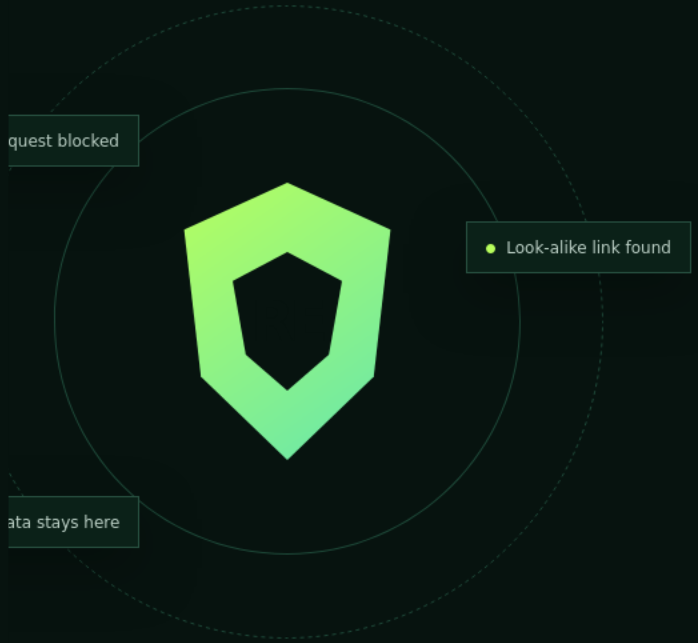
ON-DEVICE SOCIAL-ENGINEERING FIREWALL

Rakshak Edge

A private safety check before the user clicks, pays, or shares a secret



SNAPDRAGON X SERIES

Hariharan C | 2026

The critical moment happens before the tap

A suspicious message exploits urgency and trust. Sending it to a cloud checker can expose the same private content the user is trying to protect.

01

Private content

OTP messages, account alerts, payment handles, and personal context.

02

Fast manipulation

Threats and rewards push the user to act before checking another channel.

03

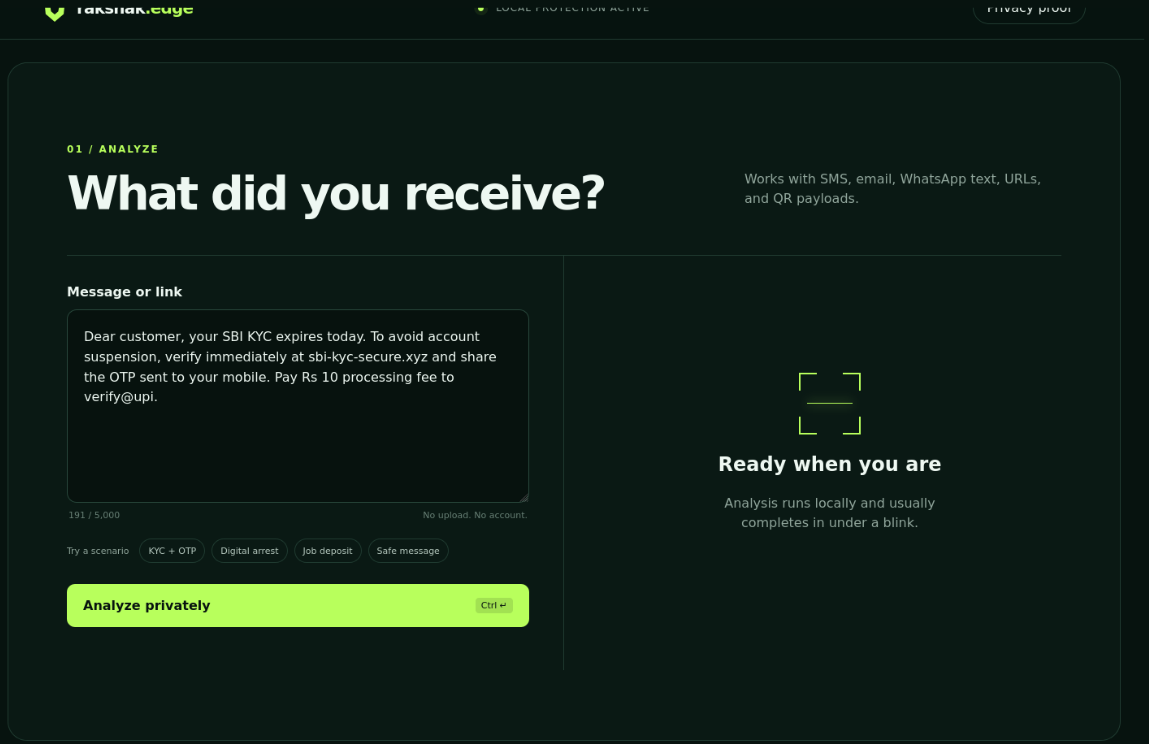
Opaque verdicts

A score alone cannot teach the user what triggered the warning or what to do next.

Design question

Can a PC explain the risk locally, in the moment that matters?

One local workflow from suspicion to safe action



Paste

Message, URL, UPI request, or QR payload

Understand

Risk score with visible evidence and extracted entities

Act safely

Independent verification path without another risky link

Hybrid reasoning keeps security explainable

Transparent signals stay authoritative. A compact classifier adds semantic context and runs on the Hexagon NPU.

01

Input boundary

Loopback API
No persistence

>

02

Evidence engines

URL + entity rules
INT8 classifier

>

03

Snapdragon path

ONNX Runtime
QNN Execution Provider

>

04

Decision layer

Calibrated score
Reasons + safe action

Graceful fallback

QNN unavailable? ONNX Runtime uses CPU. Model absent? The deterministic engine still delivers a complete, testable product.

A KYC scam becomes a teachable decision

99 /100

HIGH RISK

Stop and verify independently

OTP request	Credential theft signal	+30
Look-alike SBI domain	Identity mismatch	+26
Expires today	Artificial urgency	+18
UPI + Rs 10	Financial ask	+18

SAFEST NEXT STEP

Do not reply, pay, share a code, or open the link yet.

Privacy and responsiveness from the same architecture

0
cloud calls in the default build

INT8
target model precision

QNN
accelerator execution path

45 TOPS
Snapdragon X Elite NPU

Why the NPU matters

The transformer encoder runs without occupying the CPU or sending personal content to a data center. The rules layer remains sub-millisecond and handles crisp security facts.

Target acceptance: warm analysis below 100 ms, no outbound connection, identical label contract on CPU fallback.

A working submission, not a concept mockup

01	**Runs today**	One command, Python standard library, offline demo
02	**Tested**	Safety regression suite and reproducible latency harness
03	**Accessible**	Keyboard flow, high contrast, reduced motion, plain language
04	**Deployable**	QNN adapter, INT8 model contract, target acceptance checks
05	**Responsible**	Threat model, bounded claims, no server-side history

```
python app.py → localhost:8765
```

RAKSHAK EDGE

A safer pause before the tap

Private by design. Explainable by default. Ready for Snapdragon.

LIVE DEMO

```
python app.py
```

GitHub-ready repository | Pitch deck | Brief | Demo script

Hariharan C